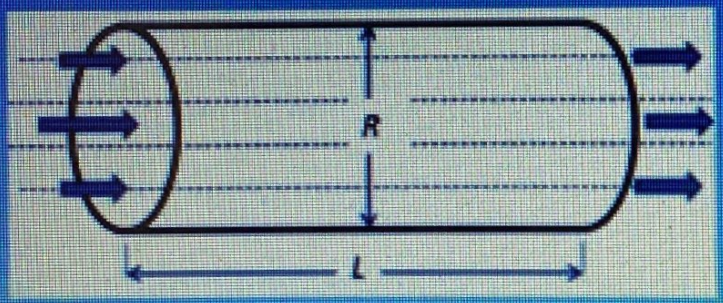
HYDRAULIC

Pipe Flow
and
Open Channel Flow

Lecture 2

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Pipe Flow

- The transport of a fluid in a closed conduit **under pressure** is called

  pipe flow

- If the absolute pressure is bigger than the atmosferic presssure the flow is called the pipe flow

$$P_{absolute} > P_{atm}$$

✓ The closed conduit is called pipe if it is round, otherwise duct.

✓ Not all conduits are round in cross-section, most of the common ones are.

WHY?

- Easy manufacture,
- Withstand a considerable pressure difference across their walls without distortion of their shapes
- The wetted perimetre is the minimum for round cross-sections for the same cross sectional areas. Minimum energy loss due to minimum friction.

Hence

"Hydraulically economical cross-sections should be designed".

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The main DRIVING FORCE in pipe flow

PRESSURE GRADIENT

(Gravitational force is ignored at the pipe lines which have mild slope)

(Pressurized Flow)

If the pipe is not full it is not possible to maintain this pressure difference

(p_1-p_2)

Some examples;

In buildings, the water pipes

Distribution system of water; pipe networks,

Circulatory system in human and animal bodies,

Irrigation,

.....

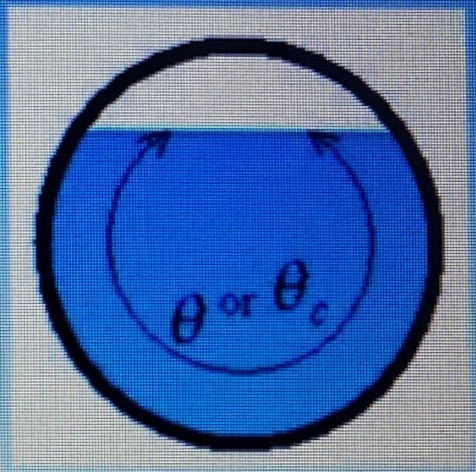
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CHANNEL FLOW (Free Surface Flow)

Even a closed conduit is not completelly filled (which means has a free surface)

or

open conduit involves the flow of a liquid is called as



OPEN CHANNEL FLOW

Examples;

Irrigation channels, drainage channels, sewerage systems, rivers

THE DRIVING FORCE

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Flow regimes through Pipes

Flow regime (mode) is important in the design and operation of any fluid system.

The amount of fluid friction which determines the amount of energy required to maintain the desired flow, depends on the mode of flow.

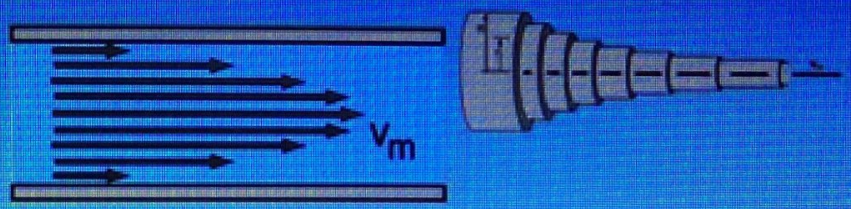
Two modes of flow:

laminar or turbulent.

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1- Laminar Flow

Laminar Flow: Fluid flows parallel layers, no distruption between layers. At low velocities the fluid tends to flow without lateral mixing, no-cross-currents no eddies, low velocities.

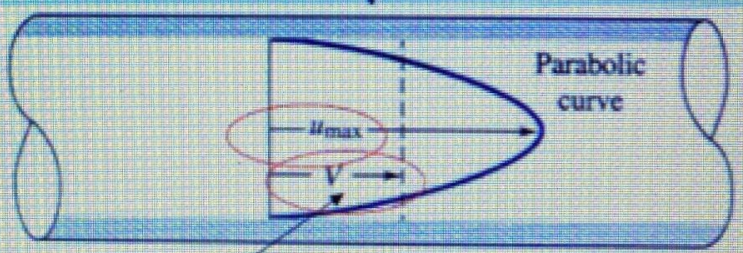


Viscous effects cause the fluid to stick to the wall.
Since the viscous effects decrease towards the axis of the pipe,

Maximum velocity is at the axis.

The velocity profile is PARABOLIC

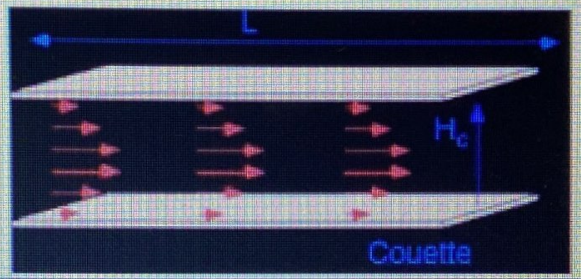
Stationary surface; the fluid in contact with the wall has zero velocity



Average Velocity

In laminar flow average velocity is 0.5 times of the maximum velocity in a pipe.

In between parallel plates (in laminar flow) ($V=0.66V_{max}$) 0.66 times of the maximum velocity.



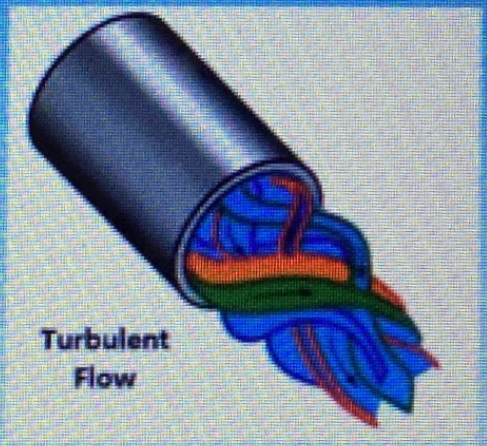
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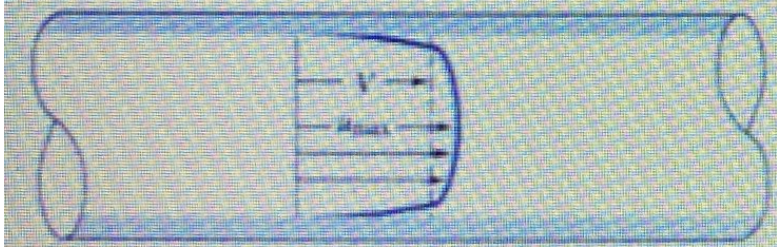
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2- Turbulent Flow

The fluid undergoes irregular fluctuations, or mixing, in contrast to laminar flow in which the fluid moves in smooth paths or layers.

In turbulent flow the speed of the fluid at a point is continuously undergoing changes in both magnitude and direction.



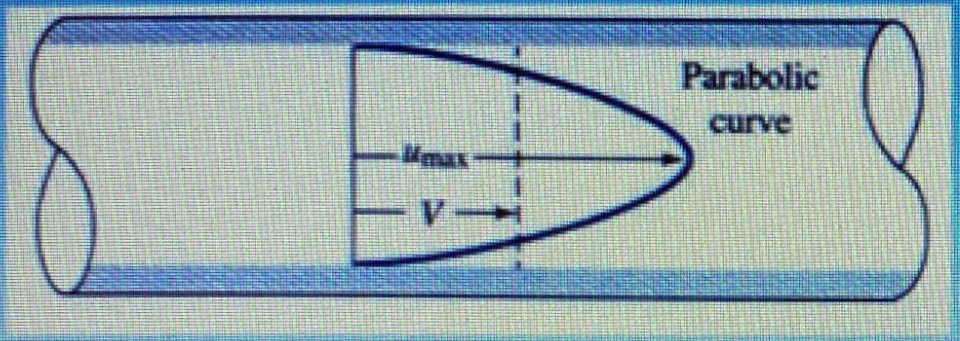


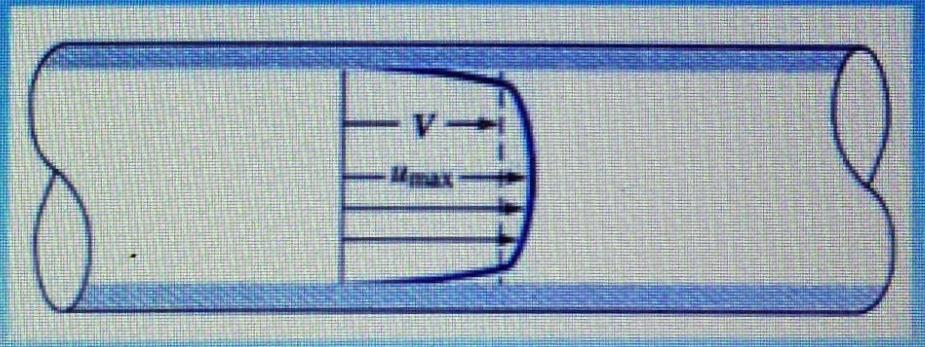
The velocity distribution in turbulent flow is more uniform and flatter than the laminar flow due to momentum exchange between fluid layers.

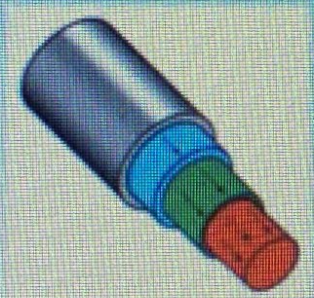
For the fully developed turbulent flow the average velocity is 0.8 times the maximum velocity.

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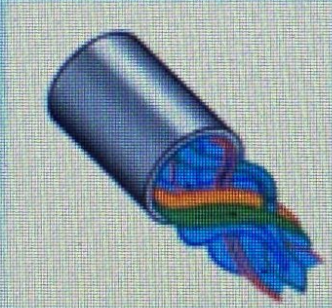
Both laminar and turbulent pipe flows produce velocity profiles that are symmetric about the axis of the pipe with a maximum velocity at the centre of the pipe.







Laminar Flow



Turbulent Flow

$$V = 0.5\,u_{max}$$

$$V = 0.8\,u_{max}$$

$$Re = VD/\nu$$

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Distribution of Velocity in Pipe Flow

The most important parameter to define is

The Average Velocity (V) in a cross section.

- Velocity distribution is related to the shear stress distribution.

∴ First of all, the shear stress distribution needs to be obtained.

How?

By using «equilibrium equation».

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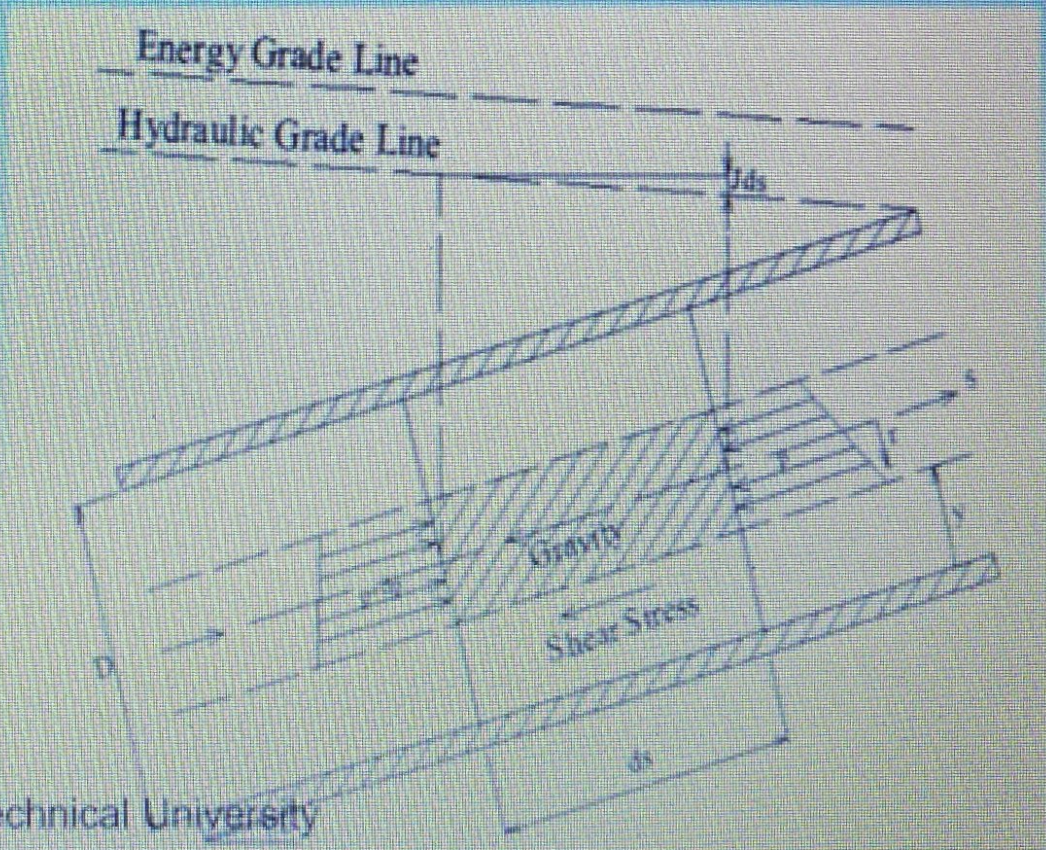
Distribution of Velocity in Pipe Flow

If we write the equilibrium of forces for a steady flow (acceleration is zero);

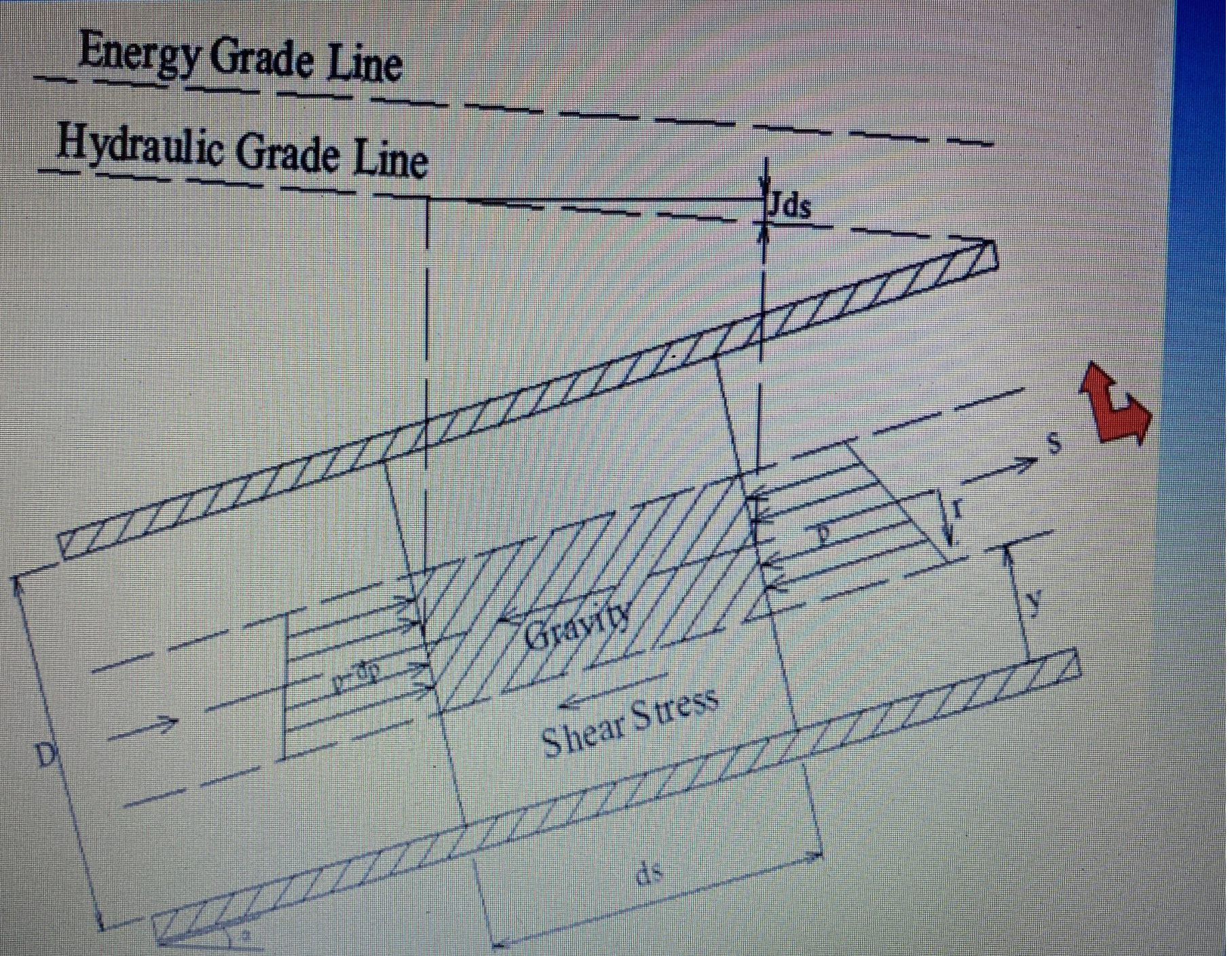
The flow is in (s) direction, Summation of forces ⟶ F=ma

The forces acting on the cylindrical fluid element are:

- Pressure against fore front and back front are different.
- gravity
- Shear along perimeter.



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If we write the equilibrium of forces for a steady flow (**acceleration is zero**):

The flow is in (s) direction, Summation of forces F=ma

Pressure Force Gravity Force Shear Force

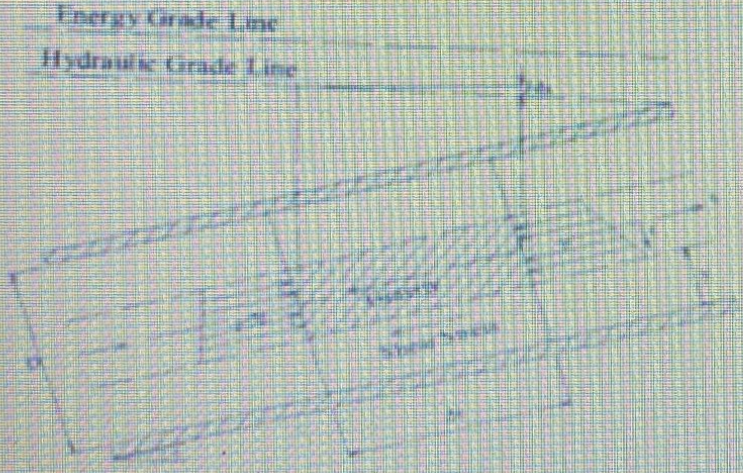
$$(p+dp)\pi r^2 - p\pi r^2 - (\rho g \pi r^2 ds)\sin\alpha - \tau_y (2\pi r ds) = 0$$

We are looking for shear stress

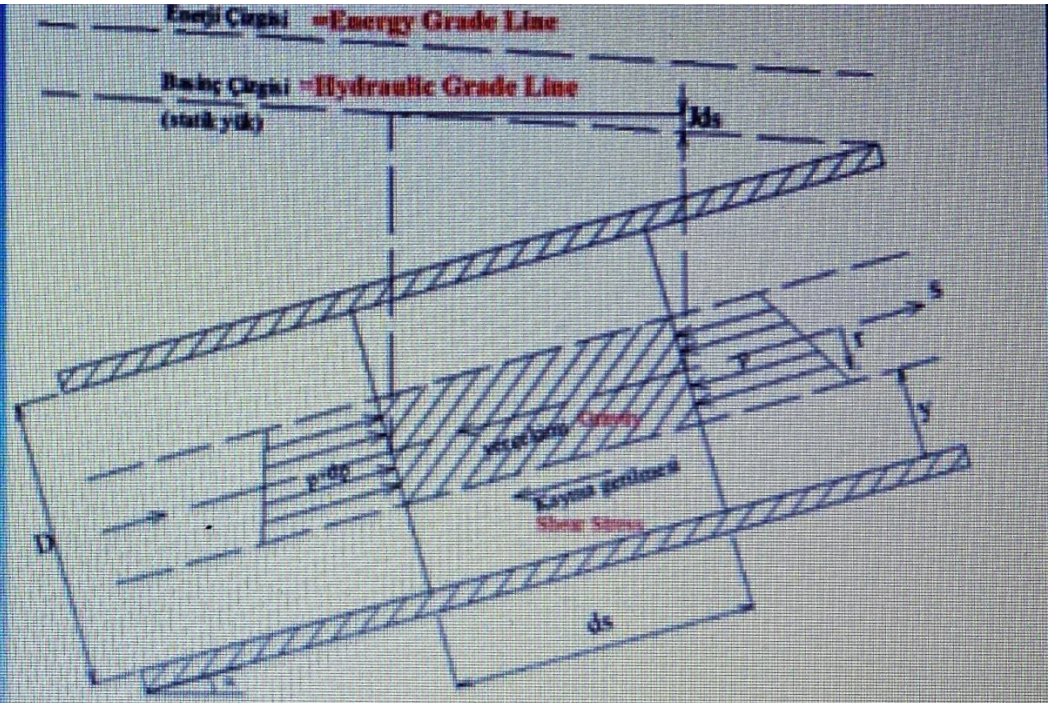
$$dp\pi r^2 - \rho g \pi r^2 ds \sin\alpha - \tau_y 2\pi r ds = 0$$

$$\left(\frac{dp}{ds} - \rho g \sin\alpha\right)\frac{r}{2} = \tau_y$$

$$\tau_y = \left(\frac{1}{\rho g}\frac{dp}{ds} - \sin\alpha\right)\rho g \frac{r}{2}$$



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Energy equation

$$\Delta h_L = \left(\frac{V^2}{2g} + \frac{(p+dp)}{\gamma} + z_1\right) - \left(\frac{V^2}{2g} + \frac{p}{\gamma} + z_2\right) = \frac{dp}{\gamma} + (z_1 - z_2)$$

$$z_2 - z_1 = ds.\sin\alpha$$

$$\Delta h_L = \frac{dp}{\rho g} - ds\sin\alpha \longrightarrow \frac{\Delta h_L}{ds} = J = \frac{1}{\rho g}\frac{dp}{ds} - \sin\alpha$$

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Pressure force Gravity Shear force

$$(p+dp)\pi r^2 - p\pi r^2 - (\rho g \pi r^2 ds)\sin\alpha - \tau_y (2\pi r ds) = 0$$

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Energy equation

$$\Delta h_L = \left(\frac{V^2}{2g} + \frac{(p+dp)}{\gamma} + z_1\right) - \left(\frac{V^2}{2g} + \frac{p}{\gamma} + z_2\right) = \frac{dp}{\gamma} + (z_1 - z_2)$$

$$z_2 - z_1 = ds.\sin\alpha$$

$$\Delta h_L = \frac{dp}{\rho g} - ds\sin\alpha$$

$$\frac{\Delta h_L}{ds} = J = \frac{1}{\rho g}\frac{dp}{ds} - \sin\alpha$$

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